L. J Institute of Engineering and Technology
List of Practical for ML and AML

Sr. No.	Practical
Decision Tree	
1.	Prepare decision tree classifier in python using sklearn library for data set diabetes.csv - (pregnant, glucose, bp, skin, insulin, bmi, pedigree, age, label). Use all features except label as independent variable. Use complete dataset for training and limit the depth of tree up to 3 levels and plot using tree.plot_tree() method.
2.	Implement a Decision Tree Classifier and compare its performance with a SVM Classifier on the iris dataset. Display the accuracy of both models.
3.	Build decision tree classifier for iris data set. One with maximum leaf nodes up to 8 and another one with minimum sample per leaf as 5. Compare accuracy of both models.
4.	Implement a Decision Tree Classifier and compare its performance with a Logistic Classifier on the diabetes dataset. Develop Python code to display the accuracy of both models.
Cross-Validation	
5.	Write a Python code to perform Leave-One-Out Cross-Validation (LOOCV) on a dataset using a Decision Tree Classifier. The dataset consists of features and labels as specified below. **Dataset Description:** • The dataset consists of 100 samples with 5 features. • The target variable is binary (0 or 1). **Instructions:** • Import the necessary libraries. • Load or create a synthetic dataset with the specified characteristics. • Implement Leave-One-Out Cross-Validation (LOOCV). • Train a Decision Tree Classifier on each training set and evaluate it on the left-out sample. • Print the accuracy for each iteration and the average accuracy across all iterations.
Naive Bayes	
6.	The breast cancer dataset is available in the sklearn.datasets module and can be loaded using load_breast_cancer(). Implement a Naive Bayes Classifier on the Breast Cancer Dataset using python's sklearn library. Assume that all features of the dataset are continuous variables and must be used for building the model. Perform the following tasks: load the data set, split data for training and testing, build a model and predict labels of test data.
Support Vector Machine (SVM)	
7.	Consider Social_Network_Ads.csv dataset - (UserID, Gender, Age, EstimatedSalary, Purchased). Use Age and EstimatedSalary as input features and Purchased as target feature. Split test data set 30% of complete dataset. Build two models of support vector classifier in python using sklearn library, one for linear and another for RBF kernel with C and gamma parameters set. Predict test labels and print test accuracy.

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8.	Develop a Support Vector Classifier to predict whether a tumor is malignant or benign based on 30 features computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. The dataset contains 569 samples with binary labels indicating tumor type (breast_cancer dataset of sklearn). Load the dataset from sklearn. Split the dataset into training and testing sets. Train the SVC model using the training data. Visualize the results.
9.	Implement an SVC model to classify iris flowers into three species (Setosa, Versicolor, and Virginica) based on their sepal and petal dimensions. The dataset contains 150 samples with four features: sepal length, sepal width, petal length, and petal width. Load the dataset from sklearn. Split the dataset into training and testing sets. Train the SVC model using the training data. Visualize the results.
Regression (Logistic and Linear)	

10.	Imagine a telecommunications company that wants to predict whether a customer will churn (leave the service) based on various features such as age, account length, and monthly charges. The company has historical data on customers, including whether they churned or not.
	Age: (in years), Account_Length: (in months), Monthly_Charges: (in dollars)
	Churn: Target variable (1 if the customer churned, 0 otherwise)
	`'Age': [25, 34, 45, 29, 50, 38, 42, 35, 48, 55],`
	`'Account_Length': [12, 24, 36, 18, 48, 30, 42, 24, 36, 60],`
	`'Monthly_Charges': [70, 90, 80, 100, 60, 80, 90, 70, 80, 100],`
	`'Churn': [0, 1, 0, 1, 0, 1, 0, 1, 0, 1]`
	Build a logistic regression model using python that can predict the probability of a customer churning by splitting 80% of given data for training. Predict targets on trained model of test data. Print accuracy.

11.	Consider Salary.csv with years_of_experience and salary. Write a python code for fitting best fit simple linear regression with independent variable years_of_experience and dependent variable salary. Use complete dataset to train model. Plot: fitted model along with trained data points, and residual plot.
Clustering (K-Means and DBSCAN)	
12.	Write a Python script that implements the Elbow Method to determine the optimal number of clusters for a generated dataset. (Generate dataset of your choice with 14 points)
13.	Use make_blobs to generate 300 samples. Analyse the data and apply KMeans clustering and generate the graph for better representation. Show what happens if you change n_clusters count from 2 to 7 (2, 3, 4, 5, 6, 7)?
14.	Generate random dataset with 35 points. Apply KMeans clustering and generate the plot with each cluster points with different colours. What is the impact of varying cluster count? Justify your understanding with suitable graphs.
15.	Develop a Python program to implement DBSCAN using the sklearn library. Use a 2D dataset (ex. make_moons) for clustering.

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16.	Generate a random dataset with 5 columns and use 3rd and 4th column as data to implement DBSCAN. Consider eps=5 and minPts=5.
Principal Component Analysis (PCA)	
17.	Write a Python program to apply Principal Component Analysis (PCA) on the Iris dataset and reduce its dimensionality to 2 dimensions. You should use sklearn's PCA implementation. After reducing the dimensions, visualize the transformed data in a 2D scatter plot, coloring the points based on their respective classes. **Dataset Description:** • The Iris dataset consists of 150 samples with 4 features. • The target variable is categorical with three classes (Setosa, Versicolor, Virginica). **Instructions:** • Import the necessary libraries. • Load the Iris dataset from sklearn.datasets. • Separate the features and the target variable. • Standardize the features (mean = 0, variance = 1). • Apply PCA to reduce the dimensionality of the dataset to 2 dimensions. • Visualize the transformed data in a 2D scatter plot, using different colors for each class.
18.	Apply Principal Component Analysis (PCA) to the Iris dataset to reduce its dimensionality from 4 to 2 dimensions. You will visualize the PCA-reduced dataset in a 2D scatter plot using Matplotlib. **Instructions:** • Load the Iris dataset. • Briefly explore the dataset to understand its structure. Print the shape of the dataset and the first few rows to get an overview of the features and target variable. • Initialize the PCA object to reduce the dataset to 2 dimensions. • Fit the PCA model to the Iris dataset and transform the dataset. • Use Matplotlib to create a scatter plot of the PCA-reduced dataset. • Color the points based on their respective species (Setosa, Versicolor, Virginica) to visualize how well the PCA has separated the different classes. • Add appropriate titles and labels to the axes of the plot. • Include a legend to indicate which colors correspond to which species.
19.	Write a python snippet to demonstrate use of PCA on 100 samples with 3 features generated randomly.
20.	Write a python snippet to demonstrate use of PCA on dataset of your choice. Also print explained variance of all principal components.
21.	Use load_iris from sklearn.datasets to load the Iris dataset. Initialize the PCA object with n_components=2 to reduce the dataset to 2 dimensions. Use matplotlib to plot the PCA-reduced dataset in 2D.

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Linear Discriminant Analysis (LDA)	
22.	Use load_wine from sklearn.datasets to load the wine dataset. Initialize the LDA object with n_components=2 to reduce the dataset to 2 dimensions. Use matplotlib to plot the LDA-reduced dataset in 2D, with each class represented by a different color.
23.	Use load_breast_cancer from sklearn.datasets to load the cancer dataset. Initialize the LDA object with one component to reduce the dataset to 1D. Use matplotlib to plot the LDA-reduced dataset.
24.	Use load_digits() from sklearn.datasets to load the handwritten numbers dataset. Initialize the LDA object with two components to reduce the dataset. Use matplotlib to plot the LDA-reduced dataset.
25.	Use load_iris from sklearn.datasets to load the Iris dataset. Initialize the LDA object with n_components=2 to reduce the dataset to 2 dimensions. Use matplotlib to plot the LDA-reduced dataset in 2D.

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